

Student
Name

Student
Number

Assessment 4: Research Report

Purpose

The purpose of this final research report is to document and communicate the outcomes and achievements of the software engineering capstone project.

Description

The research report should document and communicate the outcomes and achievements of the software engineering capstone project. This report represents the culmination of the project lifecycle, emphasising the efforts and contributions made throughout execution of the project.

The final report will integrate in-depth findings from both the research and software development efforts, encompassing the project's design, development, implementation, and evaluation phases. It will showcase innovative solutions, methodologies, and technologies employed during the software development process. Additionally, the report will present research outcomes, methodology, results, analysis, and conclusions derived from the entire project endeavour.

Consultation with Supervisor(s) and Course Coordinator(s) is recommended to determine if specific components from the interim report may be included in the final submission. The final report should adopt a scholarly academic manuscript style, showcasing depth of analysis and critical engagement with the research field. You are expected to maintain an academic tone, presenting a well-organised report that synthesizes existing knowledge, highlights research gaps, and discusses their contributions within the broader project context, encompassing both research and software development aspects.

Your submission is expected to be approximately 10,000 words (~50 pages) as a single PDF file. Any appendices should be combined to produce a single PDF file.

Note: your report will be submitted to Turnitin for plagiarism review.

A few potential templates (i.e., LaTeX and MS Word) are provided on Canvas. In both cases, remember that they are *templates* and may not be suitable out of the box. Always adapt the template to your requirements.

Marking Criteria

Content

	Absent or poor	Below average	Average	Good	Excellent	Mark
Abstract (/5) Concise summary of project, conveying the problem, solution, and conclusion.	0-1: The abstract does not effectively communicate the project's problem, solution, or conclusion. It is confusing or incomplete, making it difficult for readers to understand the project's purpose or outcomes.	2: The abstract attempts to convey the project's problem, solution, and conclusion but is overly simplistic or unclear. It may overlook key details or fail to effectively summarise the project's main components.	3: The abstract briefly outlines the project's problem, solution, and conclusion but may lack clarity or detail in certain areas. It provides a basic overview of the project's objectives and outcomes but could benefit from more depth and precision.	4: The abstract adequately captures the project's problem, solution, and conclusion in a clear and concise manner. While some details could be more refined, the overall summary provides a good understanding of the project's main elements.	5: The abstract effectively and concisely communicates the project's problem statement, solution approach, and final conclusion. It provides a clear and comprehensive overview of the project's key aspects, allowing readers to grasp the project's scope and outcomes quickly.	
Introduction/Problem Definition (/5) Brief overview of project context. Clear problem definition and explicit project aims.	0-1: The overview of project context is absent or extremely brief, lacking essential background information on the project's purpose and scope. The problem definition is unclear or entirely missing, and the project aims are not explicitly stated, making it difficult to understand the project's goals and objectives.	2: The overview of project context is limited, offering minimal background information on the project's purpose and scope. The problem definition is vaguely defined, and the project aims are somewhat implicit or unclear, requiring further clarification to understand the project's objectives.	3: The overview of project context is somewhat brief, providing basic background information on the project's purpose and scope. The problem definition is adequately defined, and the project aims are generally clear, but there may be some ambiguity or lack of specificity in articulating project objectives.	4: The overview of project context is adequate, offering sufficient background information to understand the project's relevance. The problem definition is clear and concise, and the project aims are explicitly stated, though there may be minor areas that could be further clarified.	5: The overview of project context is thorough, providing comprehensive background information on the project's purpose, scope, and relevance. The problem definition is crystal clear, and the project aims are explicitly stated, leaving no ambiguity about the project's goals and objectives.	
Literature Review (/5) Appropriate summary of prior work. Suitable background for present work.	0-1: The literature review is largely absent or extremely brief, failing to adequately summarise prior work or establish a meaningful background for the present study. The review lacks depth and does not demonstrate an understanding of relevant literature in the field.	2: The literature review offers a limited summary of prior work, with significant gaps in coverage or relevance to the present study. The background provided for the current research is somewhat unclear or insufficiently developed.	3: The literature review includes a basic summary of prior work, covering some relevant studies and concepts. It establishes a general background for the present study, though some important literature may be missing or not fully developed.	4: The literature review provides a solid summary of relevant prior work, covering key studies and concepts that inform the present study. It establishes a suitable background for the current research, demonstrating a good understanding of foundational literature in the field.	5: The literature review presents a comprehensive and insightful summary of prior work relevant to the present study. It effectively establishes the background and context necessary for understanding the significance of the current research, demonstrating a deep understanding of the field.	

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Methodology / Design (/25) Appropriate description of project methodology and design. Appropriate description of research methodology and design. Use of appropriate software engineering tools and methods.	0-5: The description of project methodology and design lacks coherency, making it difficult to understand the project's approach and objectives. The research methodology and design are not adequately explained or are absent from the submission. The use of software engineering tools and methods is minimal or inappropriate, hindering the project's development and outcomes.	6-10: The description of project methodology and design is minimal or unclear, making it challenging to grasp the project's approach effectively. The research methodology and design are briefly mentioned but lack sufficient detail or structure. There is an attempt to use software engineering tools and methods, but their application may be inconsistent or incomplete.	11-15: The description of the project methodology and design is thorough and detailed, clearly outlining the approach taken to address project objectives. The research methodology and design are meticulously explained, demonstrating a strategic and well-planned approach to gathering and analysing data. Additionally, a wide range of appropriate software engineering tools and methods are employed effectively, enhancing the project's development and outcomes.	16-20: The description of the project methodology and design is thorough and detailed, clearly outlining the approach taken to address project objectives. The research methodology and design are meticulously explained, demonstrating a strategic and well-planned approach to gathering and analysing data. Additionally, a wide range of appropriate software engineering tools and methods are employed effectively, enhancing the project's development and outcomes.	21-25: The description of the project methodology and design is thorough and detailed, clearly outlining the approach taken to address project objectives. The research methodology and design are meticulously explained, demonstrating a strategic and well-planned approach to gathering and analysing data. Additionally, a wide range of appropriate software engineering tools and methods are employed effectively, enhancing the project's development and outcomes.	
Implementation / Evaluation (/30) Description of system implementation. Detailed evaluation of project outcomes demonstrated through appropriate testing.	0-6: The system implementation description is vague or unclear, providing minimal insight into how components were developed and integrated. The evaluation of project outcomes lacks any meaningful testing efforts, failing to demonstrate an understanding of how testing contributes to project success.	7-12: The description of system implementation lacks sufficient detail and may be incomplete in certain areas. The evaluation of project outcomes demonstrates limited testing, with minimal explanation of testing methods or their relevance to project success.	13-18: The system implementation description provides basic coverage of key components and their integration. The evaluation of project outcomes includes some testing efforts but may lack depth or overlook certain aspects crucial for comprehensive assessment.	19-24: The description of system implementation is detailed, covering the major components and their integration within the project. The evaluation of project outcomes demonstrates a good level of testing, with appropriate methods used to assess the system's effectiveness and reliability.	25-30: The system implementation description is highly detailed and comprehensive, providing a thorough overview of how each component was developed and integrated. The evaluation of project outcomes is extensive, showcasing a wide range of appropriate testing methodologies used to validate system functionality and performance.	
Results / Conclusion (/15) Appropriate presentation of results, including various non-text elements. Project conclusions are well supported by the presented results.	0-3: The presentation of results is minimal or absent. There is a lack of non-text elements, making it challenging to grasp key findings. Conclusions drawn from the results are inadequately supported by the presented data, reflecting a lack of clarity or depth in	4-6: The presentation of results is lacking. There are insufficient non-text elements to enhance understanding of key findings, relying primarily on text-based descriptions. Conclusions drawn from the results are somewhat supported by the presented data, but there are significant gaps or	7-9: The presentation of results includes some non-text elements to support key findings, although the variety or effectiveness of these elements may be limited. Conclusions drawn from the results are generally supported by the presented data,	10-12: The presentation of results is solid, incorporating appropriate non-text elements that contribute to the understanding of key findings. Conclusions drawn from the results are well-supported by the presented data, demonstrating a clear	13-15: The presentation of results is outstanding, featuring a diverse range of non-text elements such as graphs, charts, tables, or visuals that effectively communicate key findings. Conclusions drawn from the results are thorough, logically derived, and strongly	

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	connecting findings to project conclusions.	inconsistencies in how findings are connected to project outcomes.	although there may be some gaps in explanation or clarity.	connection between the findings and project outcomes.	supported by the presented data, enhancing the overall clarity and impact of the project.	
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Length and Substance

	Absent or poor	Below average	Average	Good	Excellent	MARK
Quality of report and academic literacy (/10) Good report structure with formal language. Correct punctuation, sentence structure, spelling, and grammar. Concise writing is expected; excessive length will result in mark reduction	0-2: The report lacks a clear structure and may exhibit informal language or significant errors in punctuation, sentence structure, spelling, or grammar. These issues hinder comprehension and detract greatly from professionalism. May also be excessively long without justification.	3-4: The report displays a basic level of structure and uses formal language adequately. Noticeable errors in punctuation, sentence structure, spelling, or grammar affect readability. Excessive length begins to detract from clarity and focus.	5-6: The report generally follows a good structure and employs formal language appropriately. Some minor errors may be present but do not impede understanding. Slightly excessive length reduces clarity.	7-8: The report is well-structured and uses formal language effectively. Errors are minimal and do not affect professionalism. Writing is mostly concise but may be slightly excessive in length.	9-10: The report demonstrates exceptional structure and organisation, with consistently formal language. Punctuation, sentence structure, spelling, and grammar are flawless. Writing is concise and focused.	
References (/5) Appropriate reference style and correct use of citations. Complete reference section and appropriate number of references.	0-1: The reference style is largely inappropriate or consistently deviates from the chosen citation format. Citations are either missing or used incorrectly, and the reference section is incomplete or lacks an adequate number of sources to support the content presented in the document.	2: The reference style is inconsistent and may not fully adhere to the chosen citation format. Citations are used inconsistently to support arguments and ideas, resulting in a reference section that lacks completeness or contains insufficient sources.	3: The reference style is inconsistent and may not fully adhere to the chosen citation format. Citations are used inconsistently to support arguments and ideas, resulting in a reference section that lacks completeness or contains insufficient sources.	4: The reference style is mostly appropriate and consistent, with minor deviations from the chosen citation format. Citations are generally used to support arguments and ideas, contributing to a comprehensive reference section that includes an adequate number of relevant sources.	5: The reference style is consistently appropriate throughout the document, following a suitable citation format accurately. The citations are used effectively to support arguments and ideas, demonstrating a thorough and complete reference section with a suitable number of high-quality sources.	

NOTE: Submissions after the deadline will have marks reduced by **10% per day** late. (The weekend counts as 2 days.)